

Erik Nordquist

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20 Penn St, Baltimore, MD 21201

Education

- 2023 **Ph.D.** Chemistry, University of Massachusetts Amherst.
- 2018 **B.S.** Chemistry and Physics, The College of Idaho.
Majors: Chemistry, Physics. Minors: Computer Science, History, Philosophy.

Appointments

- 2023– **Postdoctoral Fellow**, University of Maryland, Baltimore, Department of Pharmaceutical Sciences.
- 2025– **Adjunct Professor**, Stevenson University, Department of Chemistry.

Fellowships and Awards

Fellowships

- 2023–26 **NIH T32 Postdoctoral Fellowship**, University of Maryland, Baltimore.
- 2022 **Teaching Fellowship**, College of Natural Sciences, University of Massachusetts Amherst.
Competitive fellowship that enabled me to design and teach a First-Year Seminar course.
- 2020–22 **NIH T32 Predoctoral Fellowship**, University of Massachusetts Amherst.

Awards and Honors

- 2026 **Graduate of the Last Decade (GOLD) Alumni**, The College of Idaho. For service to the College and outstanding professional achievements as a recent graduate.
- 2025 **Excellence in Service Award**, University of Maryland, Baltimore. For scientific outreach efforts in the local Baltimore community and service to the UMB campus community.
- 2024 **Best of Biophysical Journal**, for publication titled “Inner pore hydration free energy...”
Outstanding Talk award, Research Symposium, UM Greenebaum Comprehensive Cancer Center.
- 2022 **Paul H. Terry Endowment Award**, Chemistry Dept., University of Massachusetts Amherst.
- 2020 **William E. McEwen Poster Award**, Chemistry Dept., University of Massachusetts Amherst.
- 2018 **(Hon. Mention) “Top-Gun” TA Award**, Chemistry Dept., University of Massachusetts Amherst.

Research Experience

- 2023– **Postdoctoral Fellow**, University of Maryland Baltimore, Department of Pharmaceutical Sciences.
- **Advisor:** Alexander D. MacKerell, Jr.
 - Developing methods in computer-aided drug design (CADD).
 - Extensive collaborations with experimental groups in drug design.
- 2018–23 **Graduate Research Assistant**, University of Massachusetts, Amherst, Department of Chemistry.
- **Advisor:** Jianhan Chen; **Thesis:** Understanding Protein Structure-Function Relationships Using Simulations and Machine Learning
 - Developed free-energy simulation methods to study ion channel gating mechanisms.
 - Developed predictive models of protein function using molecular simulations and machine learning.

Teaching Experience

Stevenson University

2025– **Instructor of record**, General Chemistry Lab, Stevenson University. Full teaching responsibility for the lab class, including grading and office hours. Fall 2025, Spring 2026, Fall 2026 (ongoing).

University of Maryland, Baltimore

2020–25 **Guest lecture**, Computer-aided Drug Design in a graduate Cancer Biology course (3x);

University of Massachusetts Amherst

2022 **Instructor of record**, First-Year Seminar, Title: “Reconciling Atomic Chaos and Human Order.” I designed this course through the College of Natural Science Teaching Fellowship, including a two-day “teaching bootcamp.” ([link](#)) **Course goals**: develop scientific communication skills and perform 4-week research project. Assessments were primarily formative, with self-feedback to build growth mindsets.

Guest lecture, Discussion seminar on AlphaFold2 in a Biophysics course at Amherst College.

2020 **Guest lecture**, Graduate Statistical Mechanics lecture on molecular mechanics and simulation.

2018–19 **Graduate Teaching Assistant**, General Chemistry lab. Responsibilities: in-class instruction, grading, office hours, and exam proctoring.

Mentoring Experience

University of Massachusetts

- Samantha Schultz, undergraduate (2020–2021); now PhD student at UPenn.
- Callie Jillson, undergraduate (2019–2020); now nurse in Massachusetts.

University of Maryland

- Ruchira Basu, grad. student (2026–)
- Dami Showunmi, undergraduate, (2025–26)
- Anthony O'Donnell, grad. student (2024–)

Publications

#: Contributed equally.

Undergraduate mentees underlined.

Forthcoming:

- **Nordquist E**, MacKerell A. Automated Pharmacophore Hypothesis Generator for SILCS-Pharm: ph4gen. *J. Comput. Aid. Mol. Des.* 2026. (in prep.)
- Chan A[#], Eberly C[#], Goodis C[#], **Nordquist E**, Weldemariam M, Cooper B, Pogash S, Lapidus R, Kane M, MacKerell A, Civin C, Fletcher S. PROTAC “Linkerology” Enhances the Degradation Selectivity Profile of a Multi-CDK Inhibitor. *ACS Pharmacol. Transl. Sci.* 2026. (under review)

Published:

- **Nordquist E**, Zhao M, Yu W, MacKerell A. Computational Modeling of PROTAC Ternary Complexes as Ensembles Using SILCS-xTAC. *J. Chem. Inf. Model.* 2025. DOI: [10.1021/acs.jcim.5c02045](https://doi.org/10.1021/acs.jcim.5c02045)
- Patel R, **Nordquist E**, Polli J. Prediction of surfactant-mediated dissolution of poorly soluble drugs from drug particles. *Eur. J. Pharm. Sci.* 2025. DOI: [10.1016/j.ejps.2025.107052](https://doi.org/10.1016/j.ejps.2025.107052)
- **Nordquist E**[#], Zhao M[#], Kumar A, MacKerell A. Physics- and Machine-Learning Based Method to Identify Druggable Binding Sites Using SILCS-Hotspots. *J. Chem. Inf. Model.* 2024. DOI: [10.1021/acs.jcim.4c01189](https://doi.org/10.1021/acs.jcim.4c01189)
- **Nordquist E**, Jia Z, Chen J. Small Molecule NS11021 Promotes BK Channel Activation by Increasing Inner Pore Hydration. *J. Chem. Inf. Model.* 2024. DOI: [10.1021/acs.jcim.4c01012](https://doi.org/10.1021/acs.jcim.4c01012)
- **Nordquist E**[#], Zhang G[#], Barethiya S, Ji N, White K, Han L, Jia Z, Shi J, Cui J, and Chen J. Incorporating physics to overcome data scarcity in predictive modeling of protein function: a case study of BK channels. *PLOS Comput. Biol.* 2023. DOI: [10.1371/journal.pcbi.1011460](https://doi.org/10.1371/journal.pcbi.1011460)
- Zhang L, Barethiya S, **Nordquist E**, Chen J. Machine Learning Generation of Dynamic Protein Conformational Ensembles. *Molecules* 2023. DOI: [10.3390/molecules28104047](https://doi.org/10.3390/molecules28104047)

- **Nordquist E**, Zhiguang J, Chen J. Inner pore hydration free energy controls activations of the big potassium channel and its mutants. **Biophys. J.** 2023. DOI: [10.1016/j.bpj.2023.02.005](https://doi.org/10.1016/j.bpj.2023.02.005)
- **Nordquist E[#]**, **Schultz S[#]**, and Chen J. Using Metadynamics To Explore the Free Energy of Dewetting in Biologically Relevant Nanopores. **J. Phys. Chem. B** 2022 DOI: [10.1021/acs.jpcc.2c04157](https://doi.org/10.1021/acs.jpcc.2c04157)
- **Nordquist E**, Clerico E, Chen J, Gierasch L. Computational Modeling of Hsp70-Client Interactions: Past, Present, and Future. **J. Phys. Chem. B** 2022. DOI: [10.1021/acs.jpcc.2c03806](https://doi.org/10.1021/acs.jpcc.2c03806)
- **Nordquist E**, English C, Clerico E, Sherman W, Gierasch L, Chen J. Physics-based modeling provides predictive understanding of selectively promiscuous substrate binding by Hsp70 chaperones. **PLOS Comput. Biol.** 2021. DOI: [10.1371/journal.pcbi.1009567](https://doi.org/10.1371/journal.pcbi.1009567)
- Gong X, Chiricotto M, Liu X, **Nordquist E**, Feig M, Brooks C, Chen J. Accelerating the generalized born with molecular volume and solvent accessible surface area implicit solvent model using graphics processing units. **J. Comput. Chem.** 2020. DOI: [10.1002/jcc.26133](https://doi.org/10.1002/jcc.26133)

Presentations

Conference Presentations

- 2025 **Talk**, Frontiers at the Chemistry-Biology Interface Symposium, Johns Hopkins University, "Computational modeling of ternary complexes as ensembles using SILCS-PROTAC," Baltimore, MD.
- 2024 **Talk**, Institute for Bioscience and Biotechnology Research, University of Maryland Baltimore, Early-Career Research Symposium, "Computational design of PROTACs", Rockville, MD.
Talk, University of Maryland Greenbaum Comprehensive Cancer Center Research Symposium, "Physics- and machine-learning-based method for identifying druggable binding sites with SILCS-Hotspots." (Talk award) Baltimore, MD.
Poster, Biophysical Society Annual Meeting, "Computational mapping of allosteric modulators of the BK channel." Philadelphia, PA.
- 2023 **Poster**, Biophysical Society Annual Meeting, "A predictive model of voltage gating of BK channels via physical modeling and machine learning." San Diego, CA.
- 2022 **Poster**, Biophysical Society Annual Meeting, "Free energy of hydrophobic dewetting in gating of BK channels." San Francisco, CA.
- 2020 **Talk**, Northeastern Structural Symposium, "Physical origins of selective promiscuity to Hsp70s revealed through physics-based modeling." Virtual.
Poster, University of Massachusetts Amherst ResearchFest (WE McEwen Award): "Physical origins of selective promiscuity to Hsp70s revealed through physics-based modeling." Amherst, MA.
- 2019 **Talk**, Biophysics at University of Massachusetts Amherst, "Understanding the origins of DnaK's selective promiscuity with physics-based modeling"
Poster, Molecular Biophysics in the Northeast, "Understanding the origins of DnaK's selective promiscuity with physics-based modeling." Boston, MA.

Institutional Seminars

- 2026 **Invited Seminar**, University of Illinois, Springfield, "A Bio-Airlock: Hydration Free-Energy Controls an Ion Channel's Gating," Springfield, IL.
- 2025 **Invited Seminar**, The College of the Holy Cross, "A Bio-Airlock: Hydration Free-Energy Controls an Ion Channel's Gating," Worcester, MA.
Invited Seminar, Chemistry Department Seminar, George Washington University, "Computational design of PROTACs", Washington D.C.
- 2022 **Invited Seminar**, The College of Idaho Natural Science Symposium, "Computer simulations of proteins help understand their function." Caldwell, ID.

Talk, University of Massachusetts Amherst Chemistry Dept.: “Predicting protein function with physics, experiments and machine learning.” Amherst, MA.

Service

- 2025 **Facilitator**, Skills for Faculty Positions panel, University of Maryland Office of Postdoctoral Scholars
Moderator, Career Panel Discussion, University of Maryland, Baltimore Cancer Retreat
Poster judge, Frontiers at the Chemistry-Biology Interface Symposium, Johns Hopkins University
- 2023–24 **Facilitator**, Responsible Conduct of Research NIH training, University of Maryland, Baltimore
- 2024– **Organization committee, Annual Cancer Research Retreat**, University of Maryland, Baltimore (2x)
- 2020–20 **Organization committee, Alumni Networking Symposium**, Chemistry-Biology Interface program, University of Massachusetts Amherst (2x)
- 2020 **Search committee**, Grad. Program Manager, Chemistry Dept. University of Massachusetts Amherst
- 2019–21 **ResearchFest committee** for Chemistry Dept., University of Massachusetts Amherst (3x)

Journals Refereed for:

Biophys. J., J. Phys. Chem., ACS Omega, J. Chem. Theory Comput., J. Chem. Inf. Model., Comm. Chem., Drug Discov. Today, ACS Chemical Biology, PROTEINS, PLOS Comput. Biol.

Outreach

- 2023-26 **Volunteer instructor**, Baltimore Underground Science Space (BUGSS) ([link](#)), STEM outreach for Baltimore community members in community-accessible lab space
Volunteer instructor, RAMP Program for STEM activity for high-schoolers in Baltimore ([link](#)) (2x)
Poster judge, Mount Royal Middle School Science Fair, Baltimore (3x)
Volunteer instructor, CURE Program University of Maryland, Baltimore, STEM outreach for middle- and high-schoolers in Baltimore ([link](#)) (3x)
Volunteer, Halloween STEM outreach event at Community Engagement Center, University of Maryland, Baltimore (2x)
- 2020–23 **Reviewer for Journal of Emerging Investigators**, 25 articles by middle- / high-school students ([link](#))
- 2022 **Lab for girls summer science camp**, Eureka! at University of Massachusetts Amherst ([link](#))

Professional Development

Society Memberships:

- 2025– American Chemical Society
- 2022– Biophysical Society
- 2024 **Writing an Effective Teaching Philosophy Statement**, online CIRTL workshop
Teaching Biophysics at a PUI workshop, Biophysical Society Annual Meeting
- 2024 **Safety preparedness trainings**, CPR/AED, Stop the Bleed, Civilian Active Shooter Events, Fire Extinguishers, and Naloxone Training, University of Maryland, Baltimore ([certificates](#))
- 2022 **CITRL associate certification**, University of Massachusetts Amherst, achieved through training on evidence-based and inclusive teaching practices ([link](#))
- 2021 **Evidence-based Undergraduate STEM Teaching**, online course ([link](#))
Inclusive STEM Teaching, online course ([link](#))

Book Chapters

- **Nordquist E**, Horrigan F, MacKerell A. Computational ligand binding site prediction. **Springer Nature**. 2026. DOI: [10.1007/978-3-032-07511-6_7](https://doi.org/10.1007/978-3-032-07511-6_7)